

Supervised Machine Learning

Pharma: Drug Shelf-Life Prediction

📄 Dataset Link:

https://github.com/bhaumik2025/DS_Data/blob/main/Datasets/Supervised%20ML/Pharmaceutical%20Supply%20Chain%20Optimization.zip

📄 Tasks:

1. Load and understand dataset (columns, types, target variable).
2. Handle missing values and outliers.
3. Perform EDA (visualize distributions, correlations).
4. Feature engineering (e.g., dosage to volume ratio, temperature bands).
5. Encode categorical variables.
6. Build a Linear Regression model.
7. Build and evaluate Random Forest Regressor.
8. Compare with Gradient Boosting model.
9. Perform hyperparameter tuning using GridSearchCV.
10. Evaluate final model with RMSE and cross-validation.